

γ_{PrEP} Payer-Mix Sensitivity Analysis: Meeting Follow-Up and Revised-Parameter Results

August 13, 2026

Abstract

Part I of this memo reproduces the August 10, 2026 meeting follow-up (`INFORM-HIV-Analysis/Raff_tmp/gam`), which identified that the model's PrEP private-insurance-floor parameter ($r = 0.78$, national) is likely too high for INFORM-HIV's population (young Black MSM/TGW in Chicago) and proposed a working value of $r \approx 25\%$. Part II presents the results of actually running that revised value through the cost-mapping pipeline: a point-estimate comparison against the current committed baseline, and a small Monte Carlo sensitivity sweep over r and the baseline PrEP coverage anchor P^0 .

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Part I

Meeting Follow-Up (2026-08-10)

1 Jonathan’s question: why is it $P - \gamma$, not $P \times \gamma$?

The coverage model (main text, Eqs. 1–2) is:

$$P_{\text{ART}} = \min\left(1, \frac{G_{\text{ART}}}{\alpha C_{\text{ART}} N_{\text{HIV}+}} + \gamma_{\text{ART}}\right), \quad (1)$$

$$P_{\text{PrEP}} = \min\left(1, \frac{G_{\text{PrEP}}}{\beta C_{\text{PrEP}} N_{\text{HIV-}}} + \gamma_{\text{PrEP}}\right). \quad (2)$$

Jonathan’s question concerned the calibration equations (supplement, Eqs. 4–5), which solve for α and β given an observed baseline coverage P^0 :

$$\alpha = \frac{G_{\text{ART}}}{(P_{\text{ART}}^0 - \gamma_{\text{ART}}) C_{\text{ART}} N_{\text{HIV}+}}, \quad (3)$$

$$\beta = \frac{G_{\text{PrEP}}}{(P_{\text{PrEP}}^0 - \gamma_{\text{PrEP}}) C_{\text{PrEP}} N_{\text{HIV-}}}. \quad (4)$$

Why $(P^0 - \gamma)$ and not $(P^0 \times \gamma)$?

1.1 The subtraction is algebra, not a separate modeling choice

Equations (3)–(4) are not an independent assumption. They are what you get by taking the *forward* model, Eqs. (1)–(2), and solving for α (or β) at the observed baseline. Concretely, for ART:

$$\begin{aligned} P_{\text{ART}}^0 &= \frac{G_{\text{ART}}}{\alpha C_{\text{ART}} N_{\text{HIV}+}} + \gamma_{\text{ART}} \\ \implies \underbrace{P_{\text{ART}}^0 - \gamma_{\text{ART}}}_{\text{government-attributable share}} &= \frac{G_{\text{ART}}}{\alpha C_{\text{ART}} N_{\text{HIV}+}} \\ \implies \alpha &= \frac{G_{\text{ART}}}{(P_{\text{ART}}^0 - \gamma_{\text{ART}}) C_{\text{ART}} N_{\text{HIV}+}}. \end{aligned}$$

The subtraction appears because the forward model is **additive**: total coverage is a private floor γ (assumed fixed, unaffected by government spending) *plus* a government-driven increment $G/(\alpha CN)$, capped at 1. Isolating “how much of the observed coverage is attributable to government spending” is therefore just P^0 minus the part we’ve already assumed is privately financed — a subtraction, by construction.

1.2 The modeling choice that actually matters

The real assumption is one level up: coverage is additive (private floor + government increment), not multiplicative (e.g., government funding scaling up access among the population *not* already privately covered, or some other interaction form). We chose the additive form for parsimony; its limitations (nonlinearity, insurance-stratum heterogeneity, threshold effects) are flagged in the main text as directions for future refinement.

1.3 Why this means α, β are not free parameters

A direct consequence, and the reason this matters for the meeting’s discussion: once γ and the observed baseline P^0 are fixed, α (or β) is *determined* by Eq. (3)/(4) — it is not independently chosen or estimated. This means we **cannot revise γ_{PrEP} in isolation**: any change to γ_{PrEP} requires recalibrating β in the same step, or the model will no longer reproduce the observed baseline PrEP coverage.

1.4 A useful analytical fact: what actually drives cut-sensitivity

It’s worth separating two questions:

- **Question 1 (coverage level):** what is $P_{\text{PrEP,obs}}^0$ for young Black MSM/TGW in Chicago? (Current model: 36%, national. Evidence suggests 20–24% may be more appropriate for this population.)
- **Question 2 (payer mix):** what fraction r of PrEP users in this population have private/non-public coverage that would survive a government funding cut? (Current model: $r = 0.78$, national, giving $\gamma_{\text{PrEP}} = r \times P^0 = 0.78 \times 0.36 \approx 0.28$.)

Define the *pass-through fraction* — the share of observed baseline coverage that would be lost under a 100% funding cut (i.e., $\Delta = 1$, so $P \rightarrow \gamma$):

$$\text{pass-through} = \frac{P^0 - \gamma}{P^0} = \frac{P^0 - rP^0}{P^0} = 1 - r.$$

For PrEP, $1 - 0.78 = 22\%$; for ART, $1 - \gamma_{\text{ART}}/P_{\text{ART}}^0 = 1 - 0.15/0.62 \approx 76\%$.

The important implication: the relative pass-through fraction depends only on r (Question 2), not on P^0 (Question 1), because $\gamma = rP^0$ by construction, so P^0 cancels out of the ratio. If we only revise the coverage-level anchor (36% $\rightarrow$ 24%) while keeping the national $r = 0.78$, the model’s relative sensitivity to funding cuts is unchanged (still $\sim 22\%$ pass-through). Only the absolute number of percentage points at risk would shrink. Part II of this memo verifies this claim numerically.

2 Summary of the 2026-08-10 meeting

2.1 The problem, restated

- $\gamma_{\text{PrEP}} = 0.28$ was derived entirely from national data: a national baseline PrEP coverage of 36% (CDC Monitoring 2022) and a national payer-mix estimate that 78% of

PrEP users are privately insured (CDC NHBS 2024 / Bonacci et al. 2023).

- INFORM-HIV simulates a specific population — young Black MSM/TGW in Chicago — for whom both numbers are likely wrong in the same direction: NHBS puts Black MSM PrEP use at $\sim 24\%$ nationally (vs. 57% for White MSM), and Black MSM are known to rely far more on Medicaid/Ryan White/ADAP and less on private insurance than PrEP users nationally.
- If $r = 0.78$ is too high for this population, the model currently overstates how much PrEP access survives a funding cut, and therefore *understates* the harm of PrEP funding cuts specifically for the population the paper is about.

2.2 What we checked, and what we found

- **John Peller’s (AFC) email**, forwarded by Anna, estimates 32–33% of Illinois residents are covered by ERISA-governed self-insured employer plans, 23–25% by state-regulated fully insured plans, 35–36% by public programs, and 6–7% are uninsured. This addresses a different policy channel (removal of the USPSTF Grade A PrEP mandate) rather than the government-funding-reduction channel our model represents, but is a useful sanity-check bound: combined commercial coverage state-wide is only $\sim 55\text{--}58\%$, well below the 78% assumed among PrEP users — consistent with, but not proof of, $r = 0.78$ being too high for a lower-income, more publicly-insured subpopulation.
- A targeted literature search (AIDSVu/PrEPVu.org, CDC MMWR 2019 23-urban-areas NHBS report, the AJPH 2021 Medicaid-expansion NHBS study covering 22 cities including Chicago, and a 2018–2022 pharmacoequity claims analysis) confirmed that **no published source crosses race/ethnicity with PrEP-payer type**. This looks like a genuine gap in the literature.
- The Black–white PrEP-use gap persists *even among the insured*, and the AJPH study found the racial gap persists regardless of Medicaid-expansion status — insurance/payer source is not the whole story; a revised r corrects the financing piece but should not be read as fully correcting for the underlying access disparity.
- Meg McElroy’s ADAP FOIA summary (Illinois-specific, uninsured vs. insured client counts, April 2024–March 2025) could serve as a partial cross-check, though it is not stratified by race.

2.3 Where the discussion landed

We agreed to set $r \approx 25\%$ as a working value:

- this is a **post-hoc recalculation** — it only touches the funding-to-coverage layer, not the underlying ABM, so no new simulation is required;
- β must be recalibrated alongside any revision to γ_{PrEP} (Section 1);
- getting an expert read from AFC (John Peller/Meg McElroy) on a defensible number would be more credible than an internal guess.

3 Action plan (from the 2026-08-10 meeting)

1. Settle on a working value/range for r (equivalently, γ_{PrEP}).
2. Recalibrate β (and audit $\gamma_{\text{ART}}/\alpha$ the same way) using Eqs. (3)–(4).
3. Send AFC a narrower, specific data request: among PrEP users who are young Black MSM/TGW in Chicago, what fraction access PrEP via private insurance vs. Medicaid/ADAP/PrEP4Illinois/uninsured?
4. **Re-run the cost-mapping layer with the revised parameters** — regenerating the forest plot, the ribbon+heatmap panel, and the scenario table — and mirror the same parameter change in INFORM-HIV-App.
5. Document the limitation explicitly in the supplementary appendix: a race/city-stratified PrEP payer-mix estimate does not appear to exist in the published literature.

Part II below carries out item 4 above as an exploratory sensitivity check (not yet the paper’s committed figures — see Section 7.1).

Part II

Sensitivity Analysis Results (2026-08-13)

4 What was run

Using `04_cost_mapping/sensitivity/` (branch `sensitivity/gamma-prep-r-p0` of `INFORM-HIV-Funding`) we recalibrated γ_{PrEP} and β_{PrEP} from a chosen (r, P^0) pair via Eqs. (3)–(4) (`R/recalibrate.R`), then queried the same stage-03 incidence surrogate and common-random-numbers machinery (`compute_scenario_draws_crn()`) the main-text forest plot uses, without rebuilding the full 101×101 grid. Three parameter sets:

- **Baseline** — current committed values: $r = 0.78$, $P^0 \approx 36\%$ (i.e. $\gamma_{\text{PrEP}} = 0.28$, $\beta_{\text{PrEP}} = 4.0$).
- **Case 1** — $r = 25\%$, P^0 unchanged ($\approx 36\%$).
- **Case 2** — $r = 25\%$, $P^0 = 24\%$ (both the payer-mix and the baseline-coverage-level anchor revised).

Two important scope notes: (1) this uses only the pure-computation modules (`parameters.R`, `irr_common_random_numbers.R`) and plain `ggplot2` output — it does **not** use `plot_forest.R/plot_ribbon` which require the RAND house-style package `randplot` that is not currently installed and will need to be requested from Pedro before the paper’s actual figures can be regenerated with the revised parameter. (2) Nothing in `cost_params.yml` or the tracked `output/` directory on `main` was touched — the paper’s currently committed $r = 0.78$ baseline is unchanged.

5 Case comparison: current vs. revised r

Table 1 reports the mean incidence rate ratio (IRR, year-10, vs. the no-cut baseline) with 95% credible intervals, for three government-funding-reduction scenarios at 10/25/40% cuts. **Case 1 and Case 2 are numerically identical** on every row (verified to floating-point precision by construction, not just approximately) — directly confirming the analytical claim in Section 1: $(P^0 - \gamma_{\text{PrEP}})/P^0 = 1 - r$ is algebraically independent of P^0 . Only r is shown in the table below as a result.

Table 1: Mean incidence rate ratio (95% CI) by scenario and funding-reduction level.

Scenario	Funding cut	Current ($r = 0.78$)	Revised ($r = 25\%$)
Reduce PrEP only	10%	1.009 [0.998, 1.020]	1.033 [1.000, 1.065]
	25%	1.024 [0.998, 1.049]	1.085 [1.017, 1.159]
	40%	1.039 [1.001, 1.076]	1.135 [1.043, 1.235]
Reduce ART only	10%	1.139 [1.108, 1.171]	1.139 [1.108, 1.171]
	25%	1.367 [1.304, 1.425]	1.367 [1.304, 1.425]
	40%	1.588 [1.498, 1.684]	1.588 [1.498, 1.684]
Reduce both PrEP and ART	10%	1.149 [1.119, 1.179]	1.176 [1.135, 1.215]
	25%	1.399 [1.338, 1.459]	1.474 [1.384, 1.573]
	40%	1.645 [1.558, 1.759]	1.775 [1.644, 1.943]

The ART-only rows are identical between columns by construction (the ART payer-mix parameter γ_{ART} was not touched); the PrEP-only and combined rows show a consistently larger incidence-rate-ratio under the revised, lower r — the effect widens as the funding cut deepens.

6 Monte Carlo sensitivity to r and P^0

To check how sensitive the headline result is to uncertainty in the revised parameters themselves, we drew $N = 200$ independent joint samples of $r \sim \text{Uniform}(0.125, 0.375)$ and $P^0 \sim \text{Uniform}(0.12, 0.36)$ (each $\pm 50\%$ around the Case 2 mode, $r = 25\%/P^0 = 24\%$), recalibrated $\gamma_{\text{PrEP}}/\beta_{\text{PrEP}}$ for each draw, and queried the single headline scenario (25% government funding cut, both ART and PrEP) for each.

- IRR ranged from 1.457 to 1.491 across the 200 draws (mean 1.474).
- $\text{cor}(\text{IRR}, r) = -1.000$ — an almost perfectly linear relationship; lower r (more Medicaid/ADAP-reliant payer mix) means a larger funding pass-through and a higher incidence rate ratio.
- $\text{cor}(\text{IRR}, P^0) = -0.016$ — statistically and practically zero, the empirical confirmation of the algebraic claim that P^0 does not affect the incidence/IRR outcomes.

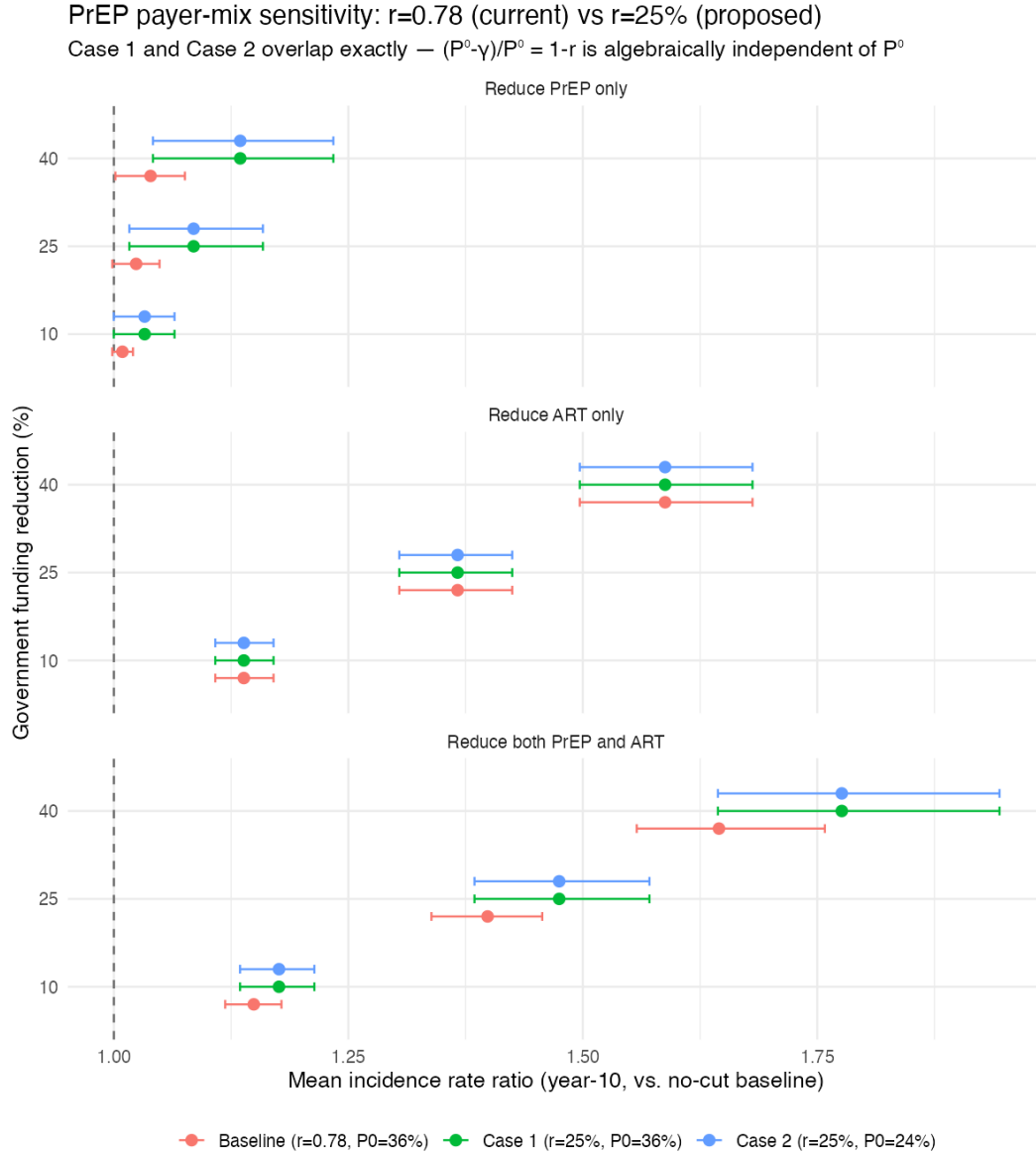


Figure 1: Forest-style comparison of mean incidence rate ratio across the three funding-cut scenarios, current ($r = 0.78$) vs. revised ($r = 25\%$, both P^0 variants). Case 1 and Case 2 overlap exactly.

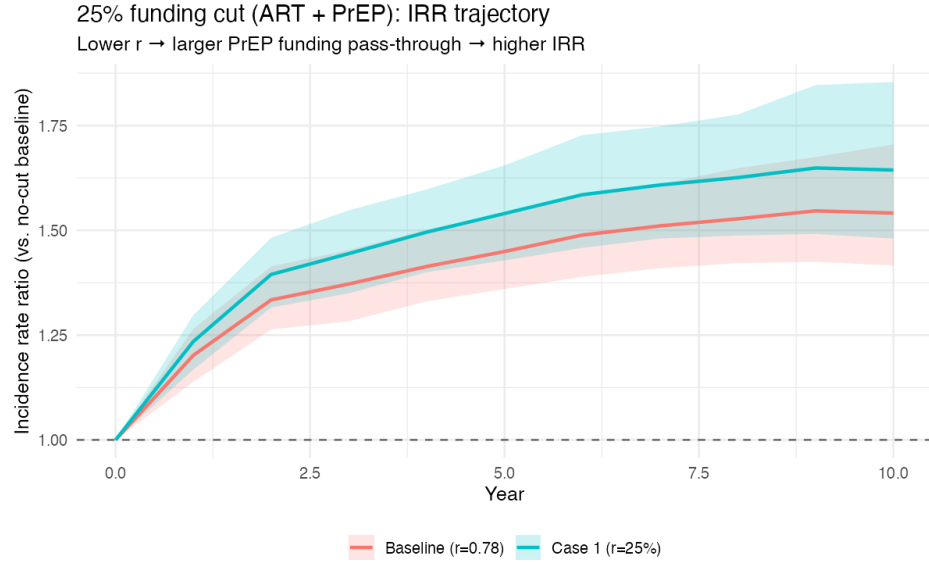


Figure 2: Year-by-year incidence-rate-ratio trajectory for a 25% government funding cut applied to both ART and PrEP simultaneously, current vs. revised r .

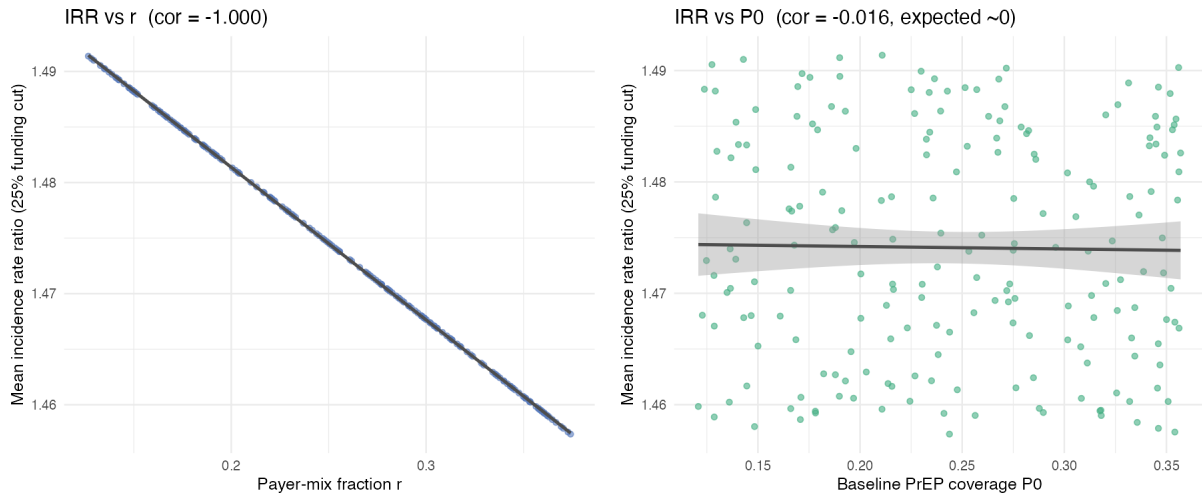


Figure 3: Monte Carlo sensitivity ($N = 200$ draws): incidence rate ratio at a 25% funding cut vs. r (left, strong negative relationship) and vs. P^0 (right, no relationship).

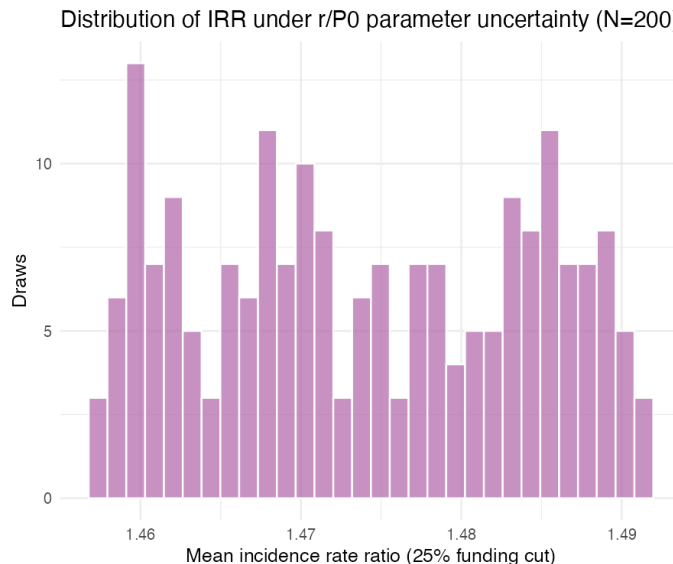


Figure 4: Distribution of the 25%-funding-cut incidence rate ratio across the 200 (r, P^0) draws.

7 Discussion

The revised, population-appropriate payer-mix assumption ($r \approx 25\%$ rather than the national $r = 0.78$) produces a materially larger projected harm from PrEP funding cuts than the paper’s currently committed baseline, consistent with the meeting’s hypothesis that the national figure understates this population’s exposure. The effect is robust to a $\pm 50\%$ uncertainty band on r itself (the sign and rough magnitude of the shift do not depend on the exact working value chosen within that range), and — as anticipated analytically — entirely insensitive to the separate, still-open question of the baseline PrEP coverage-level anchor P^0 . This means the single highest-value next step from the memo’s action plan (Section 3, item 3) is getting AFC’s read on a defensible r ; refining P^0 alone would not change the funding-cut conclusions.

7.1 Status

This is an exploratory sensitivity check, not a revision to the paper. The committed baseline on **main** ($r = 0.78$) is unchanged. Before the revised parameter can be adopted in the paper’s actual figures (which use the RAND house style via `randplot`), we need: (1) AFC’s input on a defensible r , and (2) the `randplot` package, which is not currently installed anywhere accessible and will be requested from Pedro. Item 4 of the action plan (mirroring the change into **INFORM-HIV-App**) is likewise deferred until r is actually adopted.